

Lab 2 Support Vector Machines

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Key SVM Equations

- Primal (hard-margin): $\min \|w\| \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1$
- Soft-margin: $\min \|w\| + C \sum \xi_i \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$
- Dual: $\max_\alpha \sum \alpha_i - \frac{1}{2} \sum \alpha_i \alpha_j t_i t_j K(x_i, x_j) \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \sum \alpha_i t_i = 0$
- Indicator: $\text{ind}(s) = \text{sgn}\left(\sum_i \alpha_i t_i K(x_i, s) + b\right) \Leftrightarrow \text{sgn}(w^\top \cdot \phi(s) - b)$
- Decision boundary s_0 s.t. $\sum_i \alpha_i t_i K(x_i, s_0) + b = 0$

Kernels: Linear $K = x^\top y$, Poly $(1 + x^\top y)^p$, RBF $\exp(-\|x - y\|^2 / (2\sigma^2))$

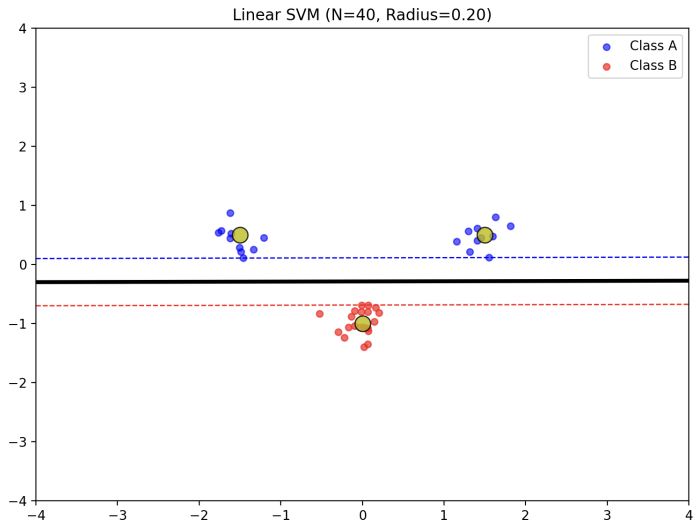
Q1: Effect of Cluster Positions and Sizes

Question: Move clusters around/change sizes to make classification easier/harder. Note when optimizer fails.

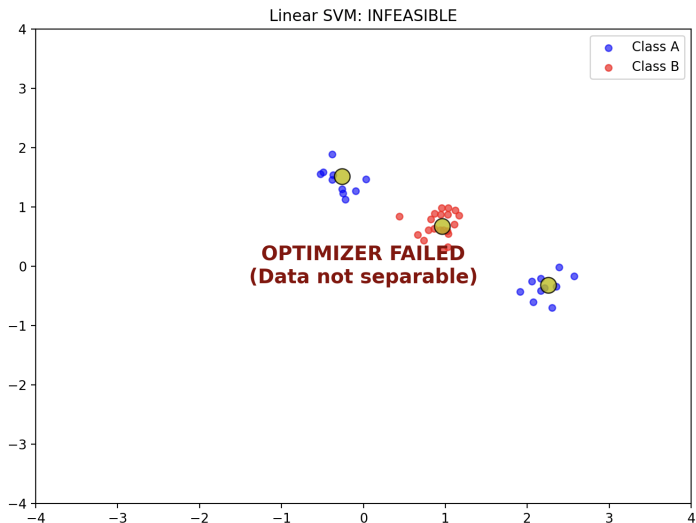
If clusters are far apart (e.g., class A at $[1.5, 0.5]$ & $[-1.5, 0.5]$, class B at $[0, -0.5]$, $\text{std}=0.2$): Easy separation, large margin, optimizer succeeds quickly. As clusters move closer or increase spread (e.g., $\text{std}=1.0$, overlap): Harder, smaller margin. If there's heavy overlap the optimizer fails for a sufficiently hard-margin / very small slack ('success'=False), as constraints can't be satisfied.

Constraints $y_i(w^\top x_i + b) \geq 1$ violated in overlap so no feasible solution exists

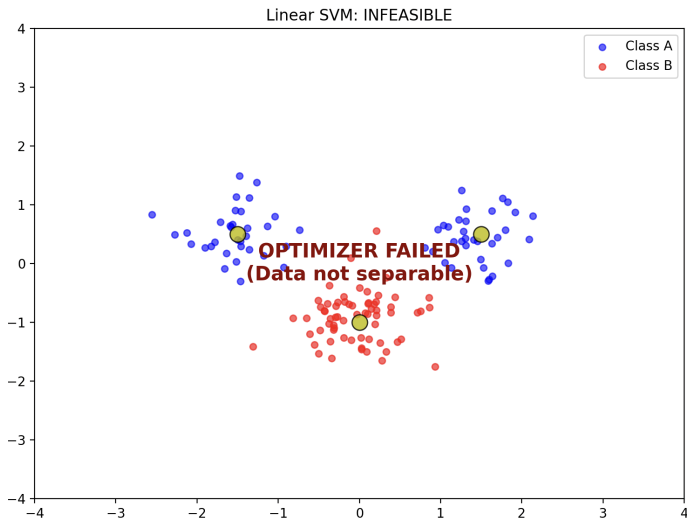
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Q2: Implementing Non-Linear Kernels

Question: Implement polynomial and RBF kernels. Classify hard datasets.

Linear kernel fails on non-linear data (e.g. circular clusters).

Polynomial: $K(x, y) = (1 + x^\top y)^p \rightarrow$ Curved boundaries (e.g. $p=2$: quadratic).

RBF: $K(x, y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \rightarrow$ Flexible, local boundaries.

- For "very hard" datasets (e.g., moon-shaped or concentric circles): RBF can separate where linear/poly fail. - Dual remains the same; just swap K in

objective: $\frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j K(x_i, x_j) - \sum \alpha_i$

Q3: Kernel Parameters & Bias-Variance Trade-off

Question: Explore how parameters influence boundary. Reason in bias-variance terms.

Polynomial degree p :

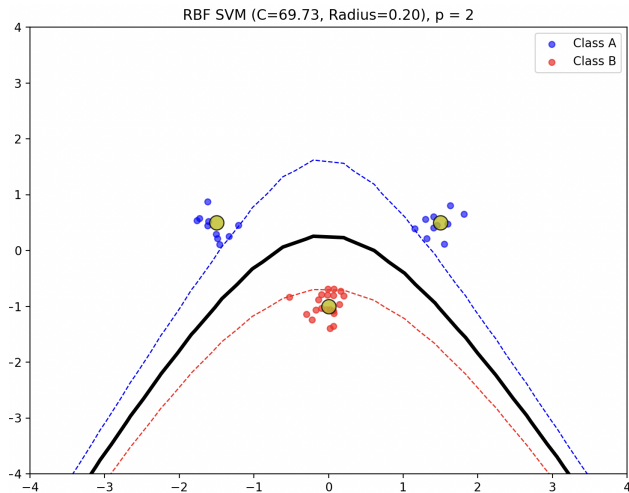
- Low p (e.g: 1=linear): High bias (underfit, simple boundary), low variance (stable).
- High p (e.g: 7): Low bias (fits complex data), high variance (overfits noise, wiggly boundary).

RBF σ (width):

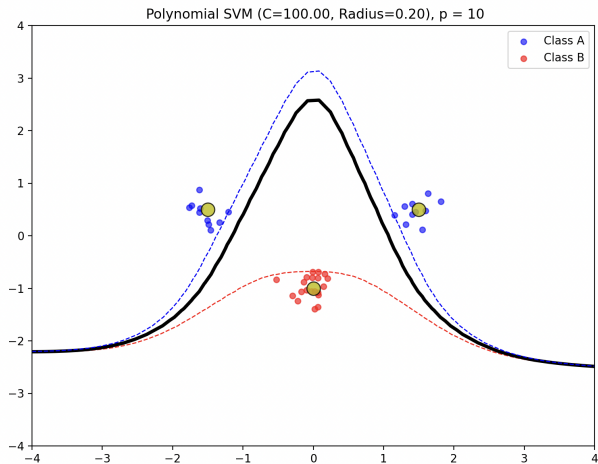
- Large σ : Smooth boundary, high bias, low variance (underfit).
- Small σ : Jagged boundary, low bias, high variance (overfit, each point influences locally).

Bias-Variance: Complex params reduce bias but increase variance. Tune to balance (e.g., via cross-validation).

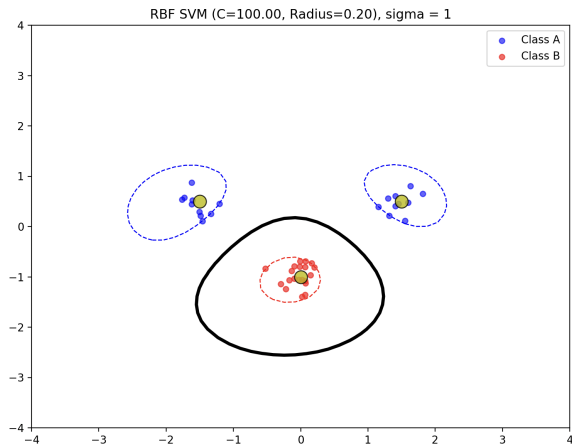
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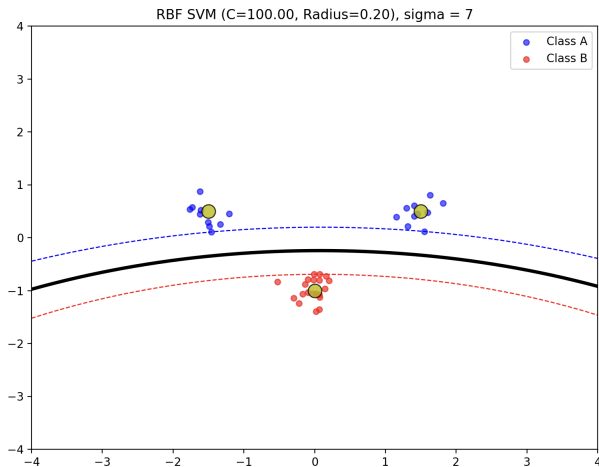
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Q4: Role of Slack Parameter C

Question: Explore C ; what for large/small values?

C controls trade-off: Margin width vs. misclassifications.

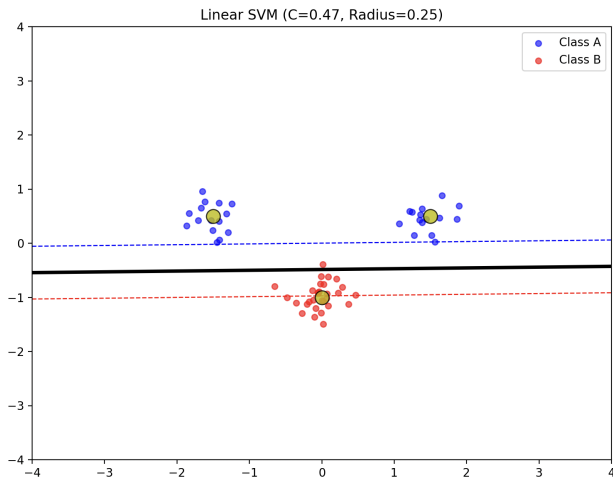
Large C (e.g: 100): Heavy penalty on ξ_i allows for almost no misclassifications of training data, small margin, potential overfit (low bias, high variance). The resulting thinner margin means the boundary decision will be more sensitive to the training data.

Small C (e.g: 0.1): Lighter penalty allows for more misclassifications of training data, larger margin, underfit if too small (high bias, low variance). The resulting thicker margin means the boundary decision means the boundary decision will be less sensitive to the training data and might generalize better.

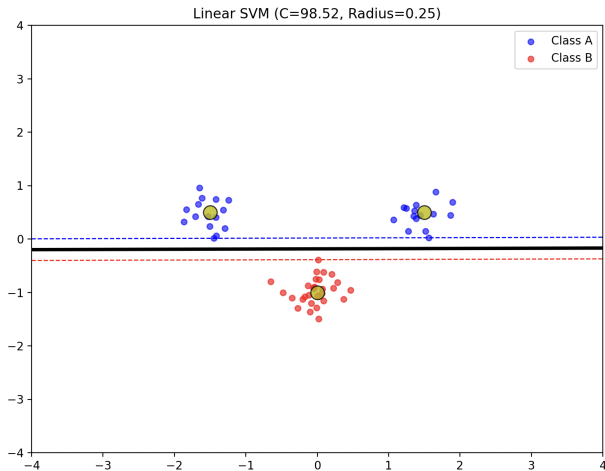
Soft-margin objective function:

$$\min \frac{1}{2} \|w\|^2 + C \sum \xi_i$$

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Q5: Slack vs. Complex Model for Non-Separable Data

Question: When opt for more slack vs. more complex kernel?

More slack (small C): When you have low training error relative to test error. Normally indicative that data has noise. Allows ignoring anomalies for larger margin, avoids overfit. Prefer when true boundary is simple but data noisy.

More complex kernel (e.g: high p poly or small σ RBF): If data inherently non-linear/complex. Captures true patterns, but risk overfit if noise present. When both your training error and test error is too high.